

MOUAD ELBAZ

Data Engineering & Machine Learning Student — Universite Sultan Moulay Slimane

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PROFESSIONAL EXPERIENCE

Data Engineering & Machine Learning Intern (PFE) — Prefecture d'Agadir Ida-Outanane, AgadirApr 15 - Jul 15, 2026

GitHub: [github.com/moad-cod/operational-intelligence-platform](#)

Designed an asynchronous ELT pipeline using a Medallion Architecture (Bronze, Silver, Gold) to consolidate over 200,000 records from GLPI ITSM, OCS Inventory, and security logs, orchestrated with Apache Airflow and modeled with dbt. Built a multilingual Retrieval-Augmented Generation (RAG) system for enterprise IT support, combining lexical (BM25) and dense vector search via Reciprocal Rank Fusion (RRF) and Cross-Encoder re-ranking, integrated with the Groq API. Trained an XGBoost classifier for IT ticket triage and an Isolation Forest model for hardware anomaly detection, achieving approximately 86% accuracy across both. Exposed all services through a FastAPI backend paired with a Next.js dashboard for real-time incident tracking and KPI visualization.

Tools: Python, Apache Airflow, dbt, PySpark, Qdrant, FAISS, XGBoost, Scikit-learn, FastAPI, Next.js, Docker, Groq API

Developer Intern — Prefecture d'Agadir Ida-Outanane, Agadir

Apr - Jun 2025

Engineered a secure, enterprise-grade attendance management system with role-based access control (RBAC), multi-tier authentication, dynamic session management, and IP compliance analysis. Implemented persistent structured audit logging for compliance verification and configured secure local networking infrastructure to restrict data exposure.

Tools: Backend Security, RBAC, Authentication, Audit Logging, Network Security, Linux

PROJECTS

RAGForge — AI Knowledge & RAG Platform — Personal Project

GitHub: [github.com/moad-cod/RAGForge](#)

Built an end-to-end SaaS application for automated document ingestion, hybrid keyword/semantic search, and metadata reranking for LLM-powered contextual Q&A. Architected relational and vector storage backends combining PostgreSQL and Qdrant, and containerized the full microservices cluster with Docker for reproducible deployment.

Tools: Python, FastAPI, PostgreSQL, Qdrant, Docker, RAG, LLMs

PlantGuard AI — Plant Disease Detection System — Academic Project

GitHub: [github.com/moad-cod/Plant-Disease-Classification-using-CNN](#)

Trained and benchmarked three architectures (a hand-built CNN, ResNet, and EfficientNet-B3) across 38 disease categories on the 70,000+ image PlantVillage dataset, reaching approximately 99.7% F1-score. Combined supervised classification with unsupervised anomaly detection (MSP and GOAD) for out-of-distribution detection, and deployed the model as an interactive Streamlit web app containerized with Docker.

Tools: Python, PyTorch, TorchVision, EfficientNet-B3, ResNet, CNN, GOAD, OpenCV, Streamlit, Docker

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, Git

Data Engineering: PySpark, Apache Hadoop, ETL/ELT, Data Pipelines, Data Lakehouse, Medallion Architecture, Data Modeling, Data Orchestration, dbt, Apache Airflow

Databases & Vector Search: PostgreSQL, MySQL, Redis, HBase, Qdrant, FAISS

Machine Learning & AI: Pandas, NumPy, SciPy, Machine Learning, Retrieval-Augmented Generation (RAG), Semantic Search, Scikit-learn

Backend & APIs: FastAPI, REST API Development, Next.js

DevOps & Cloud: Docker, Linux, System Administration, Network Security, Oracle Cloud Infrastructure (OCI)

EDUCATION

Bachelor in Big Data — Universite Sultan Moulay Slimane, Morocco

2025 - 2026

Technicien Specialise en Reseaux et Systemes — ISTA Agadir, Morocco

2023 - 2025

Baccalaureat Sciences Physiques — Lycee Qualifiant, Morocco

2022 - 2023

CERTIFICATIONS

Machine Learning Specialization (DeepLearning.AI) • Network Security (Cisco) • Microsoft Certified: Developer & Cybersecurity Pathways (Microsoft) • Introduction to Data Warehousing (IBM / Coursera)

LANGUAGES

Arabic: Native **French:** Intermediate **English:** Fluent